

Installing and Configuring Postfix & Dovecot

Tom Goncin
August 3rd, 2026
ENSE 353
Lab 6

Table of Contents

OBJECTIVE (What did we do?)	3
STEPS (How did we do it?)	4
Part 1: Postfix Installation & Global Configuration	4
Part 2: Initial Email Testing	9
Part 3: Advanced Postfix Configuration (master.cf)	10
Part 4: Dovecot Installation & Core Configuration	13
Part 5: Connecting Postfix & Dovecot (LMTP & Auth)	15
Part 6: Apache Installation and Virtual Hosts	17
Part 7: Finalizing Aliases and SMTP Testing	18
OUTCOME (Why did we do it?)	20

OBJECTIVE

(What did we do?)

To architect and secure the infrastructure of a fully functional Linux-based mail server on Rocky Linux server.

This task involves deploying **Postfix** to behave as the *Mail Transfer Agent (MTA)*
— handling *SMTP* traffic and routing.

Alongside configuring **Dovecot** as the secure *IMAP/POP3* server
— to store and manage incoming messages.

Furthermore, this lab enforces modern security standards by:

- Integrating *SSL/TLS* certificates
- Delegating user authentication
- Configuring strict firewall policies
- Generating a 4096-bit Diffie-Hellman key exchange to protect email data across the network

STEPS (How did we do it?)

Part 1: Postfix Installation & Global Configuration

1. Log in remotely as the root user, upgrade system packages and reboot the machine
`dnf upgrade --refresh`
2. Install the Postfix Mail Transfer Agent by running
`dnf install postfix`
3. Start the service with
`systemctl start postfix`
enable it to run on boot with
`systemctl enable postfix`
, and verify its status using
`systemctl status postfix`

```
[root@ed489-ense04 html]# ~systemctl start postfix
-bash: ~systemctl: command not found
[root@ed489-ense04 html]# systemctl start postfix
[root@ed489-ense04 html]# systemctl enable postfix
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' -> '/usr/lib/systemd/system/postfix.service'.
[root@ed489-ense04 html]# systemctl status postfix
• postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-03 20:40:41 CST; 25s ago
 Invocation: d95fc82cf0044ce89e49cec35610d5df
   Main PID: 7906 (master)
     Tasks: 3 (limit: 47892)
    Memory: 2.2M (peak: 4M)
       CPU: 280ms
    CGroup: /system.slice/postfix.service
            └─7906 /usr/libexec/postfix/master -w
               └─7907 pickup -l -t unix -u
                  └─7908 qmgr -l -t unix -u

Aug 03 20:40:41 ed489-ense04.engg.uregina.ca systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Aug 03 20:40:41 ed489-ense04.engg.uregina.ca postfix[7904]: postfix/postlog: starting the Postfix mail system
Aug 03 20:40:41 ed489-ense04.engg.uregina.ca postfix/postfix-script[7904]: starting the Postfix mail system
Aug 03 20:40:41 ed489-ense04.engg.uregina.ca postfix/master[7906]: daemon started -- version 3.8.5, configuration /etc/postfix
Aug 03 20:40:41 ed489-ense04.engg.uregina.ca systemd[1]: Started postfix.service - Postfix Mail Transport Agent.
```

4. Navigate to the configuration directory using
`cd /etc/postfix/`
, create a backup using
`cp main.cf /root/origconfigs/`
, and open the primary configuration file with
`emacs main.cf`

5. Set the main server hostname by uncommenting the

```
#myhostname = host.domain.tld
```

line and changing it to

```
myhostname = mail.tom.ursse.org
```

```
# INTERNET HOST AND DOMAIN NAMES
#
# The myhostname parameter specifies the internet hostname of this
# mail system. The default is to use the fully-qualified domain name
# from gethostname(). $myhostname is used as a default value for many
# other configuration parameters.
#
myhostname = mail.tom.ursse.org
#myhostname = virtual.domain.tld

# The mydomain parameter specifies the local internet domain name.
# The default is to use $myhostname minus the first component.
# $mydomain is used as a default value for many other configuration
# parameters.
```

6. Define the origin for outgoing emails by modifying the

```
#myorigin = $myhostname
```

parameter to

```
myorigin = tom.ursse.org
```

```
# SENDING MAIL
#
# The myorigin parameter specifies the domain that locally-posted
# mail appears to come from. The default is to append $myhostname,
# which is fine for small sites. If you run a domain with multiple
# machines, you should (1) change this to $mydomain and (2) set up
# a domain-wide alias database that aliases each user to
# user@that.users.mailhost.
#
# For the sake of consistency between sender and recipient addresses,
# myorigin also specifies the default domain name that is appended
# to recipient addresses that have no @domain part.
#
myorigin = tom.ursse.org
#myorigin = $mydomain
```

7. Open the network interfaces by uncommenting

```
inet_interfaces = all
```

and commenting out the option

```
localhost
```

```
# RECEIVING MAIL

# The inet_interfaces parameter specifies the network interface
# addresses that this mail system receives mail on. By default,
# the software claims all active interfaces on the machine. The
# parameter also controls delivery of mail to user@[ip.address].
#
# See also the proxy_interfaces parameter, for network addresses that
# are forwarded to us via a proxy or network address translator.
#
# Note: you need to stop/start Postfix when this parameter changes.
#
inet_interfaces = all
#inet_interfaces = $myhostname
#inet_interfaces = $myhostname, localhost
#inet_interfaces = localhost
```

8. Expand local mail delivery destinations by adding your domain

```
tom.ursse.org
```

to the front of the

```
mydestination
```

list.

```
#
# Specify a list of host or domain names, /file/name or type:table
# patterns, separated by commas and/or whitespace. A /file/name
# pattern is replaced by its contents; a type:table is matched when
# a name matches a lookup key (the right-hand side is ignored).
# Continue long lines by starting the next line with whitespace.
#
# See also below, section "REJECTING MAIL FOR UNKNOWN LOCAL USERS".
#
mydestination = tom.ursse.org, $myhostname, localhost.$mydomain, localhost
#mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
#mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain,
# mail.$mydomain, www.$mydomain, ftp.$mydomain

# REJECTING MAIL FOR UNKNOWN LOCAL USERS
#
# The local_recipient_maps parameter specifies optional lookup tables
```

9. Update the secure certificate pathways by changing the TLS file locations at the bottom of the `main.cf` file

- 9.1. Press Ctrl + s to open the search prompt.

- 9.2. Type in `smtpd_tls_cert_file` and press Enter to jump right to it.

- 9.3. You will see two lines that currently look like this:

```
smtpd_tls_cert_file = /etc/pki/tls/certs/postfix.pem
smtpd_tls_key_file = /etc/pki/tls/private/postfix.key
```

- 9.4. Change those two lines to point to your new verified certificates instead:

```
smtpd_tls_cert_file = /etc/pki/tls/certs/fullchain.pem
smtpd_tls_key_file = /etc/pki/tls/private/privkey.pem
```

```
# TLS CONFIGURATION
#
# Basic Postfix TLS configuration by default with self-signed certificate
# for inbound SMTP and also opportunistic TLS for outbound SMTP.
#
# The full pathname of a file with the Postfix SMTP server RSA certificate
# in PEM format. Intermediate certificates should be included in general,
# the server certificate first, then the issuing CA(s) (bottom-up order).
#
smtpd_tls_cert_file = /etc/pki/tls/certs/fullchain.pem
#
# The full pathname of a file with the Postfix SMTP server RSA private key
# in PEM format. The private key must be accessible without a pass-phrase,
# i.e. it must not be encrypted.
#
smtpd_tls_key_file = /etc/pki/tls/private/privkey.key
#
# Announce STARTTLS support to remote SMTP clients, but do not require that
# clients use TLS encryption (opportunistic TLS inbound).
#
smtpd_tls_security_level = may
#
# Directory with PEM format Certification Authority certificates that the
# Postfix SMTP client uses to verify a remote SMTP server certificate.
#
smtp_tls_CApath = /etc/pki/tls/certs
```

10. Done with `main.cf`

Save the file, then permit incoming SMTP traffic through the firewall using

```
firewall-cmd --permanent --zone=public --add-service=smtp
```

11. Apply the changes with

```
firewall-cmd --reload
```

and verify with

```
firewall-cmd --list-all
```

```
[root@ed489-ense04 postfix]# firewall-cmd --reload
success
[root@ed489-ense04 postfix]# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s31f6
  sources:
  services: cockpit dhcpv6-client dns http https smtp ssh
  ports: 44444/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Part 2: Initial Email Testing

1. Restart the mail server to apply the network configurations using
`systemctl restart postfix`
2. Obtain a disposable email address from temp-mail.org for safe testing. Go in web browser and type temp-mail.org
cifar85327@davopa.com was provided for me
3. Send a test email without a subject by running:
`echo "test email" | sendmail [fake_email_address]`
4. Send a formatted test email with a subject by running:
`echo -e "Subject: My test \n\n HELLO" | sendmail [fake_email_address]`
5. Verify receipt of both emails in the TempMail inbox.

```
root@ed489-ense04 postfix]# echo "test email" | sendmail cifar85327@davopa.com
You have new mail in /var/spool/mail/root
root@ed489-ense04 postfix]# echo -e "Subject: My test \n\n HELLO" | sendmail cifar85327@davopa.com
root@ed489-ense04 postfix]#
```

The screenshot shows the temp-mail.org website interface. At the top, there's a navigation bar with a logo and a link to 'tom.ursse.org'. Below that, a banner reads 'No Roaming Fees - Get eSIM from \$3'. The main content area displays 'Your Temporary Email Address' as 'cifar85327@davopa.com' with a QR code and a copy icon. Below this, a message states: 'Forget about spam, advertising mailings, hacking and attacking robots. Keep your real mailbox clean and secure. Temp Mail provides temporary, secure, anonymous, free, disposable email address.' There are four buttons: 'Copy', 'Refresh', 'Change', and 'Delete'. At the bottom, there's a red banner with the text 'Boost your productivity.' and a 'Start' button. Below the banner is a table showing the inbox.

SENDER	SUBJECT	VIEW
"Super User" root@tom.ursse.org	My test	>
"Super User" root@tom.ursse.org		>

Part 3: Advanced Postfix Configuration (master.cf)

1. While in the `/etc/postfix/` directory, back up the master configuration file using
`cp master.cf /root/origconfigs/`
and open it with
`emacs master.cf`

2. Enable the submission service by uncommenting the
`submission inet n`
line and its corresponding service.
`smtpd`

```
#tlspolicy unix - - - - - tlspolicy
# Choose one: enable submission for loopback clients only, or for any client.
#127.0.0.1:submission inet n - - - smtpd
submission inet n - n - - smtpd
  -o syslog_name=postfix/submission
# -o smtpd_tls_security_level=encrypt
# -o smtpd_sasl_auth_enable=yes
```

3. Uncomment the
`syslog_name`
`smtpd_tls_security_level=encrypt`
and
`smtpd_sasl_auth_enable=yes`
options directly beneath it.

4. Turn off wrapper mode by uncommenting and modifying the line to
`-o smtpd_tls_wrappermode=no`

```
#submissions inet n - n - - smtpd
# -o syslog_name=postfix/submissions
  -o smtpd_tls_wrappermode=no
# -o smtpd_sasl_auth_enable=yes
```

5. Lock down relay and recipient rules by uncommenting and setting

`smtpd_relay_restrictions=permit_sasl_authenticated,reject`

and

`smtpd_recipient_restrictions=permit_mynetworks,permit_sasl_authenticated,reject.`

6. Configure Postfix to hand off authentication to Dovecot by adding

`-o smtpd_sasl_type=dovecot`

`-o smtpd_sasl_path=private/auth`

```
# =====
# service type private unpriv chroot wakeup maxproc command + args
#                (yes)   (yes)   (no)   (never) (100)
# =====
smtp      inet  n       -       n       -       -       smtpd
#smtp     inet  n       -       n       -       1       postscreen
#smtpd    pass  -       -       n       -       -       smtpd
#dnsblog  unix  -       -       n       -       0       dnsblog
#tlsproxy unix  -       -       n       -       0       tlsproxy
# Choose one: enable submission for loopback clients only, or for any client.
#127.0.0.1:submission inet n - n - - smtpd
submission inet n - n - - smtpd
  -o syslog_name=postfix/submission
  -o smtpd_tls_security_level=encrypt
  -o smtpd_sasl_auth_enable=yes
  -o smtpd_tls_wrappermode=no
  -o smtpd_sasl_type=dovecot
  -o smtpd_sasl_path=private/auth
# -o smtpd_tls_auth_only=yes
# -o local_header_rewrite_clients=static:all
# -o smtpd_reject_unlisted_recipient=no
#   Instead of specifying complex smtpd_<xxx>_restrictions here,
#   specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
#   here, and specify mua_<xxx>_restrictions in main.cf (where
#   "<xxx>" is "client", "helo", "sender", "relay", or "recipient").
# -o smtpd_client_restrictions=
# -o smtpd_helo_restrictions=
# -o smtpd_sender_restrictions=
# -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
# -o smtpd_recipient_restrictions=permit_mynetworks,permit_sasl_authenticated,reject
# -o milter_macro_daemon_name=ORIGINATING
# Choose one: enable submissions for loopback clients only, or for any client.
#127.0.0.1:submissions inet n - n - - smtpd
#submissions inet n - n - - smtpd
# -o syslog_name=postfix/submissions
# -o smtpd_tls_wrappermode=no
```

7. Save the file and run

```
systemctl restart postfix
```

Part 4: Dovecot Installation & Core Configuration

1. Install the Dovecot email server software by running

```
dnf install dovecot
```

2. Start and enable the service using

```
systemctl start dovecot  
systemctl enable dovecot
```

```
[root@d489-ense04 postfix]# dnf install dovecot  
Last metadata expiration check: 1:14:42 ago on Mon 03 Aug 2026 08:40:15 PM.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing: dovecot	x86_64	1:2.3.21-19.el10_2.1	appstream	4.8 M

```
=====
```

Transaction Summary

Install 1 Package

Total download size: 4.8 M
Installed size: 18 M
Is this ok [y/N]: y
Downloading Packages:
dovecot-2.3.21-19.el10_2.1.x86_64.rpm

		7.2 MB/s	4.8 MB	00:00
Total		5.7 MB/s	4.8 MB	00:00

```
=====
```

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Running scriptlet: dovecot-1:2.3.21-19.el10_2.1.x86_64
Installing : dovecot-1:2.3.21-19.el10_2.1.x86_64
Running scriptlet: dovecot-1:2.3.21-19.el10_2.1.x86_64

	1/1
Preparing	1/1
Installing	1/1
Running scriptlet	1/1

```
=====
```

Installed:
dovecot-1:2.3.21-19.el10_2.1.x86_64

Complete!
[root@d489-ense04 postfix]# systemctl start dovecot
[root@d489-ense04 postfix]# systemctl enable dovecot
Created symlink '/etc/systemd/system/multi-user.target.wants/dovecot.service' -> '/usr/lib/systemd/system/dovecot.service'.

3. Navigate to the directory using

```
cd /etc/dovecot/
```

, backup the config with

```
cp dovecot.conf /root/origconfigs/
```

, and open it with

```
emacs dovecot.conf
```

4. Modify the protocols line to

```
protocols = imap pop3 lmtp
```

(removing 'submission' so Postfix handles it).

```
# Protocols we want to be serving.  
protocols = imap pop3 lmtp
```

5. Navigate to the sub-directory using

```
cd /etc/dovecot/conf.d
```

, backup

```
cp 10-mail.conf /root/origconfigs/
```

, and open it in the editor.

6. Define the file format and directory by uncommenting

```
mail_location = maildir:~/Maildir
```

```
# See doc/wiki/Variables.txt for full list. Some examples:
#
mail_location = maildir:~/Maildir
# mail_location = mbox:~/mail:INBOX=/var/mail/%u
# mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
```

7. Grant Linux system group permissions by uncommenting and setting

```
mail_privileged_group = mail
```

```
# Group to enable temporarily for privileged operations. Currently this is
# used only with INBOX when either its initial creation or dotlocking fails.
# Typically this is set to "mail" to give access to /var/mail.
#mail_privileged_group = mail
```

8. Save the file and add the dovecot user to the mail group by executing

```
gpasswd -a dovecot mail
```

```
bash: emacs: command not found
[root@ed489-ense04 conf.d]# emacs 10-mail.conf
[root@ed489-ense04 conf.d]# gpasswd -a dovecot mail
Adding user dovecot to group mail
[root@ed489-ense04 conf.d]#
```

Part 5: Connecting Postfix & Dovecot (LMTP & Auth)

1. While in the `conf.d` directory, back up and open `10-master.conf`
2. Locate the `service lmtp` block and modify the `unix_listener` to specifically target `/var/spool/postfix/private/dovecot-lmtp`, changing the mode to `0600` and setting both user and group to `postfix`.

```
service lmtp {  
    unix_listener /var/spool/postfix/private/dovecot-lmtp  
    {  
        mode = 0600  
        user = postfix  
        group = postfix  
    }  
}
```

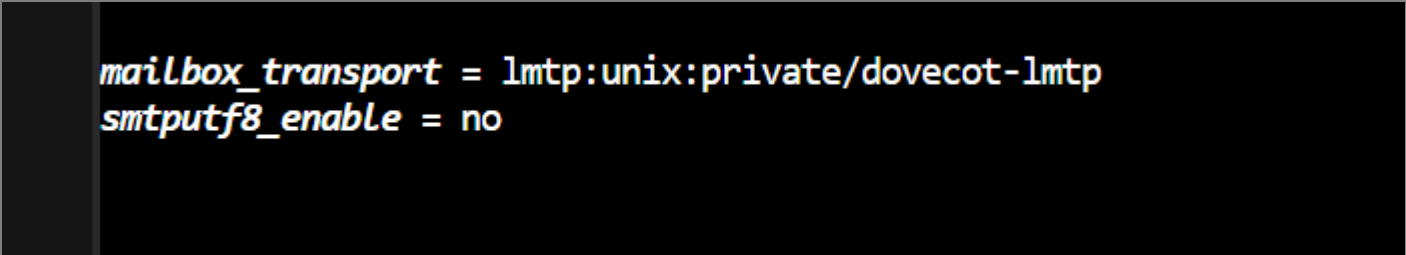
3. Locate the `Postfix smtp-auth` block and modify the `unix_listener` to target `/var/spool/postfix/private/auth`, changing the mode to `0600` and setting the user and group to `postfix`.

```
# Postfix smtp-auth  
unix_listener /var/spool/postfix/private/auth  
{  
    mode = 0600  
    user = postfix  
    group = postfix  
}
```

4. Save the file, then navigate back to `/etc/postfix/` and open `main.cf`

5. At the bottom of `main.cf`, append

```
mailbox_transport = lmtp:unix:private/dovecot-lmtp
and
smtputf8_enable = no
```

A terminal window with a black background and light blue text. It shows the configuration changes being made to the postfix main.cf file. The first line is 'mailbox_transport = lmtp:unix:private/dovecot-lmtp' and the second line is 'smtputf8_enable = no'.

```
mailbox_transport = lmtp:unix:private/dovecot-lmtp
smtputf8_enable = no
```

6. Restart both services using

```
systemctl restart postfix
systemctl restart dovecot
```

Part 6: Apache Installation and Virtual Hosts

1. Navigate back to `/etc/dovecot/conf.d`, backup and open `10-auth.conf`

2. Uncomment

```
disable_plaintext_auth = yes
```

and modify the username format to

```
auth_username_format = %n
```

3. Update the mechanisms line to

```
auth_mechanisms = plain login
```

4. Save the file, backup and open

```
10-ssl.conf
```

5. Correct the TLS pathways by pointing `ssl_cert` to

```
</etc/pki/tls/certs/fullchain.pem
```

and `ssl_key` to

```
</etc/pki/tls/private/privkey.pem
```

6. Uncomment the lines

```
ssl_dh
```

```
ssl_min_protocol = TLSv1.2
```

7. Uncomment and modify the ciphers line to

```
ssl_prefer_server_ciphers = yes
```

8. Save the file, then generate the secure Diffie-Hellman parameters by running

```
openssl dhparam -out /etc/dovecot/dh.pem 4096
```

(Note: This process takes approximately 20 minutes)

Part 7: Finalizing Aliases and SMTP Testing

1. Navigate to the root configuration folder using `cd /etc` and open the aliases file with

```
emacs aliases
```

2. At the bottom of the file, uncomment the root mail redirection and change it to your specific system username (e.g., root: tom).

```
root: Copernicus
```

3. Restart both services again using

```
systemctl restart postfix
```

```
systemctl restart dovecot
```

4. Initiate a raw text-based network connection to the local SMTP server by typing

```
telnet 127.0.0.1 25
```

```
[root@ed489-ense04 etc]# emacs /etc/dovecot/conf.d/10-ssl.conf
[root@ed489-ense04 etc]# systemctl restart dovecot
[root@ed489-ense04 etc]# telnet 127.0.0.1.25
telnet: 127.0.0.1.25: Name or service not known
127.0.0.1.25: Unknown host
[root@ed489-ense04 etc]#
```

5. Execute the manual SMTP handshake and mail delivery using the following sequence:

- HELO mail.tom.ursse.org
- MAIL FROM: any@sender.com
- RCPT TO: [your_temp_mail_address]
- DATA
- Subject: Test Subject
- [Email Body Text]
- . (A single period to end the data block)

```
[root@ed489-ense04 etc]# telnet 127.0.0.1 25
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
220 mail.tom.ursse.org ESMTD Postfix
HELO mail.tom.ursse.org
250 mail.tom.ursse.org
MAIL FROM: any@sender.com
250 2.1.0 Ok
RCPT TO: cifar85327@davopa.com
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
Subject: Test Subject
This is the body of the test email.
.
250 2.0.0 Ok: queued as 6B0F710038AB4
QUIT
221 2.0.0 Bye
Connection closed by foreign host.
[root@ed489-ense04 etc]#
```

6. Verify the final manual email was successfully delivered to the TempMail inbox.

The screenshot shows the TempMail website interface. At the top, the browser address bar displays 'temp-mail.org'. Below the navigation bar, a banner for 'Start Your SMS Business' is visible. The main content area features a dashed box titled 'Your Temporary Email Address' containing the email address 'cifar85327@davopa.com'. To the right of the address are icons for QR code and copy. Below this, a paragraph explains the service: 'Forget about spam, advertising mailings, hacking and attacking robots. Keep your real mailbox clean and secure. Temp Mail provides temporary, secure, anonymous, free, disposable email address.' Underneath the text are four buttons: 'Copy', 'Refresh', 'Change', and 'Delete'. At the bottom, there is a banner for 'vistaprint' with the text 'Signs that turn into sales' and a 'Shop now' button. Below the banner is a table representing the email inbox.

SENDER	SUBJECT	VIEW
any@sender.com	Test Subject	>

OUTCOME

(Why did we do it?)

Upon successfully architecting the sever into a fully functioning, highly secure email infrastructure through these following key components:

MTA & SMTP Routing (Postfix):

Configured to seamlessly manage Mail Transfer Agent responsibilities and mail routing.

Storage & Retrieval Backend (Dovecot):

Integrated via Local Mail Transfer Protocol (LMTP) sockets to handle secure storage and IMAP/POP3 retrieval.

Modern Enterprise Security:

- Enforced strict TLS encryption bound to acquired SSL certificates.
- Generated a 4096-bit Diffie-Hellman key exchange for secure connections.
- Blocked unauthenticated external relays.

Verification & Testing:

Validated via manual Telnet session testing, confirming that internal routing, authentication handoffs, and external delivery rules operate successfully in tandem.